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COMMON HALF YEARLY EXAMINATION – 2025

T Standard XII Reg.No.

MATHEMATICS

Time: 3.00 hrs. Marks: 9

Part – I

I. **Choose the correct answer:** 20 x 1 = 2

1. If $\sin^{-1}x + \sin^{-1}y = \frac{2\pi}{3}$; then $\cos^{-1}x + \cos^{-1}y$ is equal to
   a) $\frac{2\pi}{3}$ b) $\frac{\pi}{3}$ c) $\frac{\pi}{6}$ d) π
2. If the function $f(x) = \sin^{-1}(x^2 - 3)$, then x belongs to
   a) $[-1, 1]$ b) $[\sqrt{2}, 2]$ c) $[-2, -\sqrt{2}] \cup [\sqrt{2}, 2]$ d) $[-2, -\sqrt{2}]$
3. If $A = \begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix}$, B = adj A and C = 3A, then $\frac{|\text{adj } B|}{|C|}$ =
   a) $\frac{1}{3}$ b) $\frac{1}{9}$ c) $\frac{1}{4}$ d) 1
4. If $x^a y^b = e^m$, $x^c y^d = e^n$, $\Delta_1 = \begin{vmatrix} m & b \\ n & d \end{vmatrix}$, $\Delta_2 = \begin{vmatrix} a & m \\ c & n \end{vmatrix}$, $\Delta_3 = \begin{vmatrix} a & b \\ c & d \end{vmatrix}$, then the values of x and y are respectively,
   a) $e^{(\Delta_2/\Delta_1)}, e^{(\Delta_3/\Delta_1)}$ b) $\log(\Delta_1/\Delta_3), \log(\Delta_2/\Delta_3)$
   c) $\log(\Delta_2/\Delta_1), \log(\Delta_3/\Delta_1)$ d) $e^{(\Delta_1/\Delta_3)}, e^{(\Delta_2/\Delta_3)}$
5. The equation of the normal to the circle $x^2 + y^2 - 2x - 2y + 1 = 0$ which is parallel to the line $2x + 4y = 3$ is
   a) $x + 2y = 3$ b) $x + 2y + 3 = 0$ c) $2x + 4y + 3 = 0$ d) $x - 2y + 3 = 0$
6. The area of the triangle formed by the complex numbers z, iz and z+iz in the Argand's diagram is
   a) $\frac{1}{2}|z|^2$ b) $|z|^2$ c) $\frac{3}{2}|z|^2$ d) $2|z|^2$
7. The equation of the circle passing through the foci of the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ having centre at (0, 3) is
   a) $x^2 + y^2 - 6y - 7 = 0$ b) $x^2 + y^2 - 6y + 7 = 0$
   c) $x^2 + y^2 - 6y - 5 = 0$ d) $x^2 + y^2 - 6y + 5 = 0$
8. If α, β and γ are the zeros of $x^3 + px^2 + qx + r$, then $\sum \frac{1}{\alpha}$ is
   a) $-\frac{q}{r}$ b) $-\frac{p}{r}$ c) $\frac{q}{r}$ d) $-\frac{q}{p}$
9. If $z = x + iy$ is a complex number such that $|z + 2| = |z - 2|$, then the locus of z is
   a) real axis b) imaginary axis c) ellipse d) circle
10. If a vector $\vec{\alpha}$ lies in the plane of $\vec{\beta}$ and $\vec{\gamma}$, then
    a) $[\vec{\alpha}, \vec{\beta}, \vec{\gamma}] = 1$ b) $[\vec{\alpha}, \vec{\beta}, \vec{\gamma}] = -1$ c) $[\vec{\alpha}, \vec{\beta}, \vec{\gamma}] = 0$ d) $[\vec{\alpha}, \vec{\beta}, \vec{\gamma}] = 2$
11. A random variable X has binomial distribution with n = 25 and p = 0.8 then standard deviation of X is
    a) 6 b) 4 c) 3 d) 2

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12. If $w(x, y) = x^y$, $x > 0$, then $\frac{\partial w}{\partial x}$ is equal to

a) $x^y \log x$ b) $y \log x$ c) yx^{y-1} d) $x \log y$

13. The differential equation of the family of curves $y = Ae^x + Be^{-x}$, where A and B are arbitrary constants is

a) $\frac{d^2y}{dx^2} + y = 0$ b) $\frac{d^2y}{dx^2} - y = 0$ c) $\frac{dy}{dx} + y = 0$ d) $\frac{dy}{dx} - y = 0$

14. The area between $y^2 = 4x$ and its latus rectum is

a) $\frac{2}{3}$ b) $\frac{4}{3}$ c) $\frac{8}{3}$ d) $\frac{5}{3}$

15. Angle between $y^2 = x$ and $x^2 = y$ at the origin is

a) $\tan^{-1}\frac{3}{4}$ b) $\tan^{-1}\left(\frac{4}{3}\right)$ c) $\frac{\pi}{2}$ d) $\frac{\pi}{4}$

16. In the last column of the truth table for $\neg(p \vee \neg q)$ the number of final outcomes of the truth value 'F' are

a) 1 b) 2 c) 3 d) 4

17. If $\int_0^x f(t)dt = x + \int_x^1 tf(t)dt$, then the value of f (1) is

a) $\frac{1}{2}$ b) 2 c) 1 d) $\frac{3}{4}$

18. Subtraction is not a binary operation in

a) R b) Z c) N d) Q

19. If $f(x, y) = e^{xy}$ then $\frac{\partial^2 f}{\partial x \partial y}$ is equal to

a) $xy\, e^{xy}$ b) $(1 + xy)\, e^{xy}$ c) $(1 + y)\, e^{xy}$ d) $(1 + x)\, e^{xy}$

20. The solution of the DE $y\frac{dx}{dy} = \cot x$ is _____.

a) $\sec x = cy$ b) $\sec y = cx$ c) $\sec y = c$ d) $\sec x = c$

Part - II

II. **Answer any 7 questions. (Q.No.30 is compulsory)** 7 x 2 = 14

21. Prove that $\tan^{-1}\frac{1}{2} + \tan^{-1}\frac{1}{3} = \frac{\pi}{4}$

22. Find the square root of $6 - 8i$.

23. If $y = 4x + c$ is a tangent to the circle $x^2 + y^2 = 9$, find c

24. Find the angle between the line $\vec{r} = (2\hat{i} - \hat{j} + \hat{k}) + t(\hat{i} + 2\hat{j} - 2\hat{k})$ and the plane $\vec{r}.(6\hat{i} + 3\hat{j} + 2\hat{k}) = 8$

25. Compute the limit $\lim_{x \to a}\left(\frac{x^n - a^n}{x - a}\right)$

26. Evaluate : $\int_0^{\pi/2} \sin^{10} x\, dx$

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27. Use the linear approximation to find approximate values of $(123)^{2/3}$
28. Form the differential equation by eliminating the arbitrary constants A and B from $y = A\cos x + B\sin x$.
29. If $\text{adj } A = \begin{bmatrix} -1 & 2 & 2 \\ 1 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$, find A^{-1}.
30. Suppose X is a binomial variate $X \sim B(5, p)$ and $2P(X = 2) = P(X = 3)$, then find p.

Part - III

III. Answer any 7 questions. (Q.No.40 is compulsory) **7 x 3 = 21**

31. Show that the equation $z^3 + 2\bar{z} = 0$ has five solutions
32. If α, β and γ are the roots of the polynomial equation $ax^3 + bx^2 + cx + d = 0$, find the value of $\sum \frac{\alpha}{\beta\gamma}$ in terms of the coefficients.
33. Find the equation of the hyperbola with vertices $(0, \pm 4)$ and foci $(0, \pm 5)$.
34. Find the equation of the plane passing through the intersection of the planes $2x + 3y - z + 7 = 0$ and $x + y - 2z + 5 = 0$ and is perpendicular to the plane $x + y - 3z - 5 = 0$.
35. The probability density function of X is given by $f(x) = \begin{cases} kxe^{-2x} & \text{for } x > 0 \\ 0 & \text{for } x \le 0 \end{cases}$. Find the value of k.
36. Find the absolute maximum and absolute minimum values of the function $f(x) = 2x^3 + 3x^2 - 12x$ on $[-3, 2]$
37. Find the volume of a sphere of radius a.
38. Show that $p \to q$ and $q \to p$ are not equivalent
39. Solve the following system of linear equations by matrix inversion method: $2x + 5y = -2$, $x + 2y = -3$
40. If $u = e^{2(x-y)}$ prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = u \log u$

Part - IV

IV. Answer all the questions. **7 x 5 = 35**

41. a) Using vector method, prove that $\cos(\alpha - \beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta$

(OR)

b) If the normal at the point 't_1' on the parabola $y^2 = 4ax$ meets the parabola again at the point 't_2', then prove that $t_2 = -\left(t_1 + \frac{2}{t_1}\right)$

42. a) If $ax^2 + bx + c$ is divided by $x + 3$, $x - 5$, and $x - 1$, the remainders are 21, 61 and 9 respectively. Find a, b and c. (Use Gaussian elimination method.)

(OR)

b) If $z = x + iy$ and $\arg\left(\frac{z-i}{z+2}\right) = \frac{\pi}{4}$, then show that $x^2 + + y^2 + 3x - 3y + 2 = 0$

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43. a) Assume that water issuing from the end of a horizontal pipe, 7. 5 m above the ground, describes a parabolic path. The vertex of the parabolic path is at the end of the pipe. At a position 2. 5 m below the line of the pipe, the flow of water has curved outward 3m beyond the vertical line through the end of the pipe. How far beyond this vertical line will the water strike the ground?

(OR)

b) Solve: $(x-5)(x-7)(x+6)(x+4) = 504$

44. a) Find the vector parametric, vector non–parametric and Cartesian form of the equation of the plane passing through the points (–1, 2, 0) , (2, 2,–1) and parallel to the straight line $\frac{x-1}{1} = \frac{2y+1}{2} = \frac{z+1}{-1}$

(OR)

b) Verify
(i) closure property
(ii) commutative property
(iii) associative property
(iv) existence of identity, and
(v) existence of inverse for following operation on the given set $m * n = m + n - mn$; $m, n \in Z$.

45. a) A police jeep, approaching an orthogonal intersection from the northern direction, is chasing a speeding car that has turned and moving straight east. When the jeep is 0.6 km north of the intersection and the car is 0.8 km to the east. The police determine with a radar that the distance between them and the car is increasing at 20 km/hr. If the jeep is moving at 60 km/hr at the instant of measurement, what is the speed of the car?

(OR)

b) The time T, taken for a complete oscillation of a single pendulum with length ℓ, is given by the equation $T = 2\pi\sqrt{\frac{\ell}{g}}$, where g is a constant. Find the approximate percentage error in the calculated value of T corresponding to an error of 2 percent in the value of ℓ.

46. a) Find the area of the region common to the circle $x^2 + y^2 = 16$ and the parabola $y^2 = 6x$. *(OR)*

b) A farmer plans to fence a rectangular pasture adjacent to a river. The pasture must contain 1,80,000 sq. mtrs in order to provide enough grass for herds. No fencing is needed along the river. What is the length of the minimum needed fencing material.

47. a) A random variable X has the following probability mass function

X	1	2	3	4	5	6
f(x)	k	2k	6k	5k	6k	10k

Find (i) $P(2 < X < 6)$ (ii) $P(2 \le X < 5)$ (iii) $P(X \le 4)$ (iv) $P(3 < X)$ *(OR)*

b) In a murder investigation, a corpse was found by a detective at exactly 8 p.m. Being alert, the detective also measured the body temperature and found it to be 70°F. Two hours later, the detective measured the body temperature again and found it to be 60°F. If the room temperature is 50°F, and assuming that the body temperature of the person before death was 98.6°F, at what time did the murder occur? [log(2.43) = 0.88789; ; (log(0.5) =–0.69315]

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